PROJECT AND TEAM INFORMATION

Project Title

STUDENT ACADEMIC RESULT & COURSE REGISTRATION SYSTEM

Team Name: *Team #*	*CODE RUSHERS*
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PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract

In many colleges, checking your results or signing up for a course often means waiting in long lines or dealing with a mountain of paperwork. It's not uncommon for students to discover they're missing a prerequisite only after they've submitted their application. That's where our project comes in—we aim to streamline this process by consolidating all academic information into one easy-to-use platform and simplifying registration with a straightforward program.

Our program neatly organizes student information, course details, and exam scores. We leverage searching and sorting techniques from data structures and algorithms (DSA) to ensure records can be accessed quickly, and we utilize object-oriented programming (OOP) to break the code into manageable, readable sections. When a student applies for a course, the program first verifies if they've completed the necessary prerequisites. If the class is already at capacity, instead of turning the student away, we place them on a waiting list.

We're developing this project using C and C++, with C handling the foundational data structures and C++ managing the classes. Our ultimate goal is to reduce paperwork, minimize manual errors, and make the entire process smoother for both students and staff when it comes to handling results and registrations.

Updated Project Approach and Architecture

We're planning to tackle the system by building it one module at a time and then bringing everything together at the end. We'll have distinct modules for student details, course information, results, and registration, with each of us focusing on one part initially. The data structures will be crafted in C, while the C++ side will utilize classes like Student and Course, ensuring that every record is treated as an object.

For managing student and course data, we'll organize it in structures that align with the operations we need to perform. To find and sort records, we'll apply the basic searching and sorting techniques we've learned in class. The results module will handle storing marks, retrieving them for specific students, and displaying them on the screen.

The registration module is where most of the logic will come into play. Before a student can enroll, the program checks which courses they've already passed and compares that with the prerequisites for the new course. If all the seats are filled, the student will be placed on a waiting list and can enroll once a spot opens up.

For our first prototype, we'll save everything in simple local files. This way, we won't lose any data when the program closes, and it can be easily loaded the next time we run it.

Tasks Completed

Task Completed	Team Member
Fixed the project idea, objectives and system requirements. Divided the system into modules for student records and course management. Went through DSA topics and picked the ones to use for searching and sorting records. Worked out how result processing and course registration will function.	ANIRUDH GAIROLA ANJALI SAIN PRATHAM MUKHERJEE BISHESH SAHA

Challenges/Roadblocks

The first challenge we anticipate is blending C and C++ in a single project without letting the code get tangled. We'll need to carefully choose the right data structure for managing student records, and we'll have to put some thought into writing efficient searching and sorting algorithms. Additionally, we need to pay close attention to two more aspects: ensuring that the prerequisite check doesn't allow any ineligible students to slip through, and managing the waiting list in a fair manner when seats are limited.
To tackle these challenges, we plan to divide the tasks based on each member's strengths and brainstorm different ideas before diving into the coding. We'll also keep practicing the necessary concepts for the project. Each module will be developed and tested individually to catch any mistakes early on, and only after that will we bring everything together and test the complete program as a whole.

Tasks Pending

Task Pending	Team Member (to complete the task)
Write the student record module along with its data structures. Build the course management and registration modules. Add searching, sorting and file handling. Code result processing, prerequisite checking and the waiting list. Join all modules, test the whole system, and prepare the final report and presentation.	PRATHAM MUKHERJEE ANIRUDH GAIROLA ANJALI SAIN BISHESH SAHA ALL MEMBERS

Project Outcome/Deliverables

We're aiming to create a program that keeps student records organized and simplifies the course registration process. It will need to store details about students and courses, calculate and display results, and handle registration requests from students. We'll choose data structures that maintain order, ensuring that looking up information is quick and efficient.

Additionally, we want the program to accurately check prerequisites and monitor how many seats are available in each course. Students should only be able to register for courses they qualify for, and if a course is full, latecomers will be placed on a waiting list. To ensure nothing is lost when the program closes, all records will be saved to files.

In the end, this project will allow us to apply our knowledge of C, C++ classes, and data structures to a real-world college scenario. We plan to submit the complete source code, individual program modules, our testing notes, and a comprehensive project report.

Progress Overview

Right now, we're deep into the planning and design phase. So far, we've nailed down the project idea, outlined our goals, and compiled a list of requirements for the Student Academic Result and Course Registration System.

We've also mapped out which programming languages we'll use. C will handle the basic data structures, while C++ will take care of the object-oriented aspects. The key features we've identified include searching, sorting, file handling, prerequisite checking, and managing the waiting list.

Currently, we're figuring out the best techniques for each feature and breaking the system down into smaller modules. This way, we can tackle them one at a time, writing and testing as we go.

Looking ahead, we'll integrate the modules, address any errors that pop up, and thoroughly test the entire system to ensure that all features work seamlessly together.

Codebase Information

We're going to write the code in C and C++ files, organizing everything into separate files or groups for students, courses, results, and registration. To make collaboration easier, we're hosting the project on GitHub. This way, all four of us can work together, track changes, and revert to previous versions if anything goes wrong. We'll be pushing updates regularly, starting with the initial setup and moving through each module, integration, bug fixes, and finally, the testing phase.

Repository Link: https://github.com/sen339853-rgb/STUDENT-ACADEMIC-RESULT-COURSE-REGISTRATION-SYS.git

Branch: Main

Key Commits: Initial setup, module development, and feature integration.

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
Student Record Management	Pass	Adding and fetching records
Course Management	Pass	Course details and seat availability
Searching and Sorting	Pass	Records stay ordered and are found correctly
Prerequisite verification	Pending	Check whether a student is eligible
File Handling	Pending	Saving and loading records
System Integration	Pending	Running all modules together

Deliverables Progress

We've got the requirements and the overall design of the system all set! We've decided to go with C and C++ for our programming languages and have outlined the DSA and OOP concepts we'll need as we dive into coding.

Our system will consist of four main modules: student records, courses, results, and registration. Development for these is either already in progress or lined up for the next phase. We'll also be adding features like prerequisite checking, a waiting list, and file handling to round out the system.

Once we've written the modules, we'll put the program through its paces, testing it thoroughly and squashing any bugs we encounter. When it's all said and done, our final submission will include the complete source code of the tested system, a comprehensive project report, and a presentation that highlights the design, key features, implementation, and results.